

Pre-work · Blood Vessels and Blood

Blood Vessels, Blood Vessel Disorders and Fetal Circulation, and Blood. Read all three Part A chapters first. Most of this is histology, so work it with a slide or an image beside you.

Module 3 · 3 of 4

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the vessel and blood chapters into mini visuals. Eight chunks. Two grids do the histology and one chain does the fetal detour.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 GRID

The three tunics

Tunic down the side. Position, tissue, and one distinguishing note across the top. Add a fourth column: is it present in a capillary, yes or no.

2 GRID

The five vessel types

Vessel down the side, in the order blood meets them. Direction, wall, and role across the top. This grid is the frame for the whole chapter.

3 SPLIT

Artery against vein, on a slide

One split. Wall thickness, lumen shape, tunica media, tunica externa, and valves. This is the pair you will be asked to tell apart, so make the giveaway column obvious.

4 GRID

The three capillary types

Capillary down the side, wall and where found across the top. Add a fourth column: what can cross it. Three rows, and the third column is the reasoning column.

5 CHAIN

Fetal circulation, placenta to placenta

One long chain with arrows. Umbilical vein, ductus venosus, right atrium, then the two ways past the lungs, and back out the umbilical arteries. Mark each shunt where it appears.

6 GRID

Fetal shunt to adult remnant

Fetal structure down the side, what it does and what it becomes across the top. Five rows. Circle the one that does not become a cord.

7 HUB

Blood, composition

One spun tube at the hub. Spoke to plasma with its three proteins, the buffy coat, and the erythrocyte layer. Write the percentage beside each layer.

8 GRID

The five leukocytes

Leukocyte down the side. Group, nucleus, granules, and defense role across the top. Order the rows most to least numerous, because that order is itself an answer.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

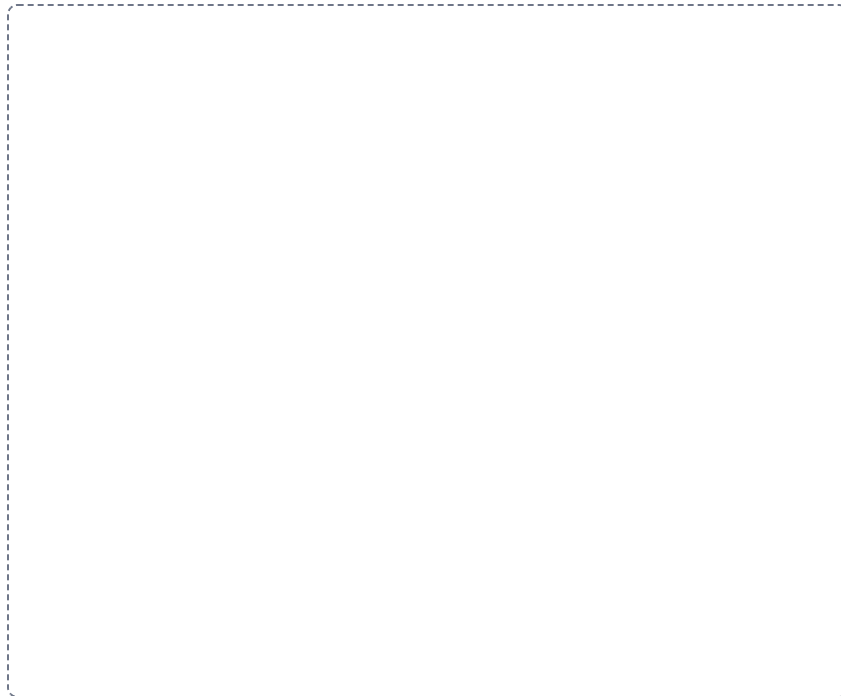
DRAWING 1

An artery and a vein, side by side in cross section

BRAIN DUMP FIRST

One minute. All three tunics in each, and every difference between them.

Draw the two vessels beside each other, cut across. Label all three tunics in each, and draw the internal and external elastic laminae where they belong. Make the artery wall thick with a round lumen and the vein wall thin with a wide, collapsed lumen. Add a valve to the vein and vasa vasorum to the externa of both.



In your notebook, head the page **M3-7 An artery and a vein, side by side in cross section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

Someone shows you an unlabelled slide and you name the vessel in under five seconds, and can say which feature decided it.

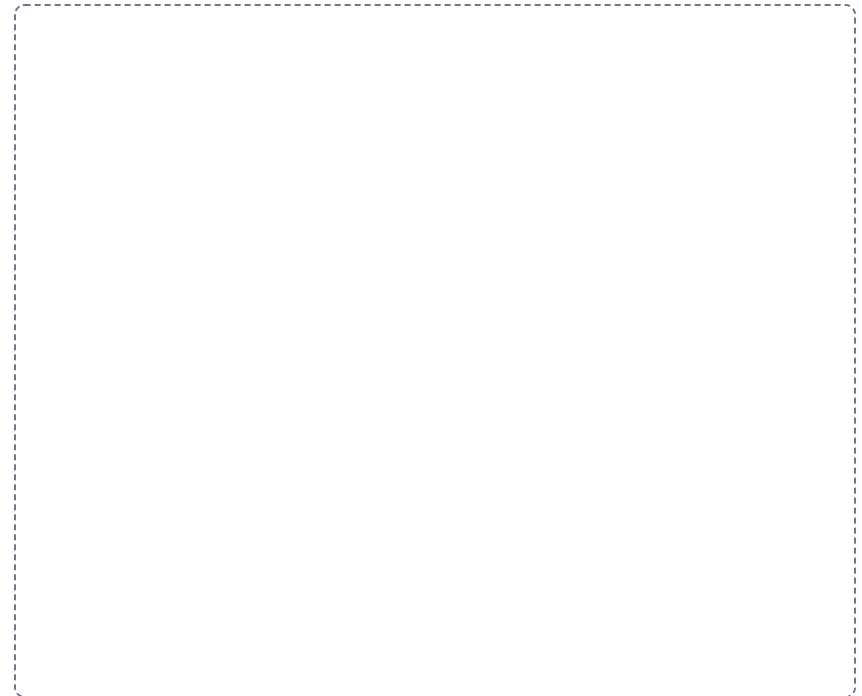
DRAWING 2

The fetal circulation, with the three shunts

BRAIN DUMP FIRST

Two minutes. Every fetal vessel and where each detour goes.

Draw a simple fetus with the heart, liver, and lungs marked, and the placenta beside it. Draw the umbilical vein in, the ductus venosus past the liver, the foramen ovale between the atria, the ductus arteriosus between the pulmonary trunk and the aorta, and the two umbilical arteries out. Arrow the whole route. Then in a second color write the adult remnant beside each shunt.



In your notebook, head the page **M3-8 The fetal circulation, with the three shunts** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain why each shunt exists by pointing at the organ it bypasses and saying what that organ is not yet doing.

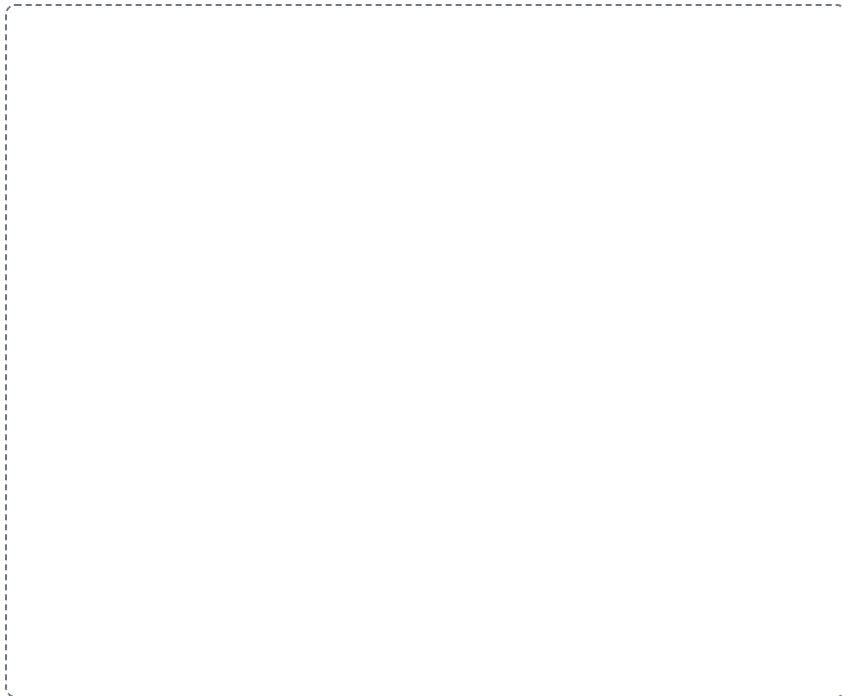
DRAWING 3

A blood smear

BRAIN DUMP FIRST

One minute. Every formed element, drawn as it looks under the microscope.

Draw a field of a blood smear. Fill most of it with biconcave erythrocytes, showing the pale center. Add one of each leukocyte, drawn by its nucleus: multi-lobed neutrophil, bilobed eosinophil, granule-obscured basophil, round-nucleus lymphocyte, kidney-shaped monocyte. Scatter platelets between them. Label each and note its relative number.



In your notebook, head the page **M3-9 A blood smear** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can identify each cell from the nucleus alone, with the labels covered.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A slide shows a vessel with a wide, irregular lumen, an incomplete basement membrane, and very large gaps between the endothelial cells. Whole blood cells are passing through the wall. Name the vessel and one organ it is found in.

Answer. A sinusoid. Liver, spleen, or red bone marrow.

Reasoning. Only one vessel in the body has an incomplete basement membrane, and that alone settles it. Work back from what is crossing: a continuous capillary keeps most substances out, a fenestrated capillary passes fluid and small solutes through pores, but whole cells need gaps far larger than either. So the tissue must be one where cells legitimately move between blood and tissue, which is exactly what the liver, spleen, and marrow do. Structure is matched to the job, and the leakiest wall goes where the largest thing has to cross.

Set A · Name it

1. The only tunic a capillary has is the _____.
2. The small vessels supplying the wall of a large vessel are the _____.
3. The ring of smooth muscle controlling entry to a capillary is the _____.
4. The adult remnant of the ductus arteriosus is the _____.
5. Name the three main plasma proteins and one job of each.

Set B · Predict it

1. A deep vein thrombosis forms in the calf and breaks free. Trace the exact route it takes and name where it lodges.

2. The valves of the leg veins fail. Predict what you would see and explain it from venous pressure.

3. The ductus arteriosus fails to close after birth. Predict the abnormal connection this leaves.

4. An abnormal hemoglobin makes a red cell rigid. Predict where in the circulation it causes trouble and why.

Set C · Tell it apart

1. Explain why blood is drawn from a vein rather than an artery, using three structural differences.

2. Distinguish an aneurysm from an atherosclerotic plaque by what each does to the vessel wall.

3. Blood is called a connective tissue. Defend that classification using its matrix.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

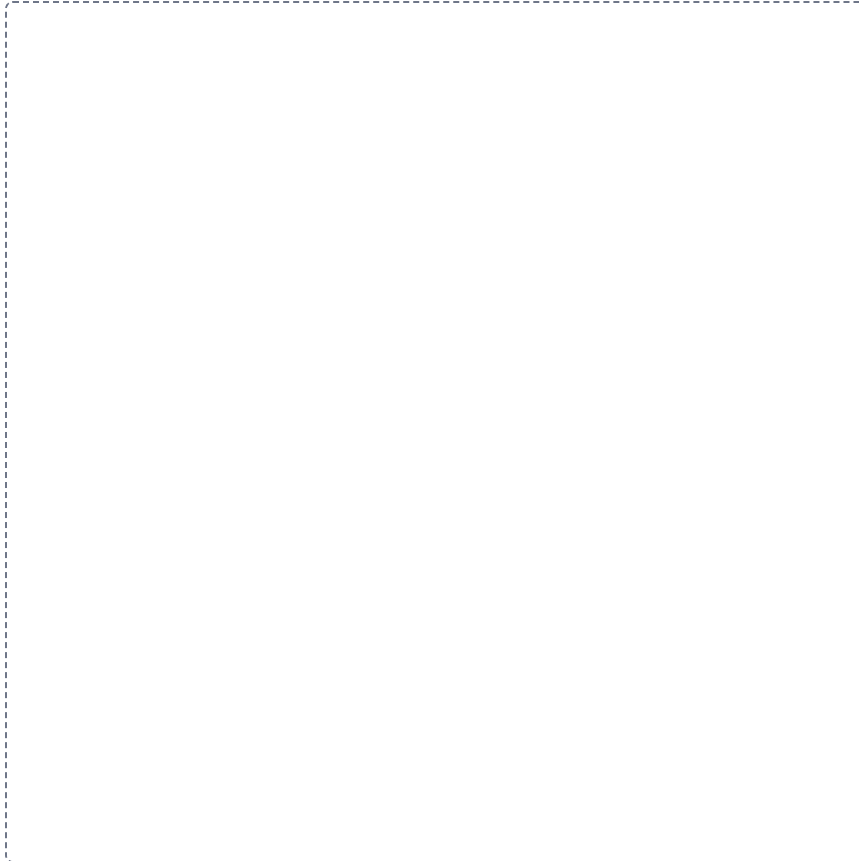
Mini visual sheet**2 min**

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

The three tunics

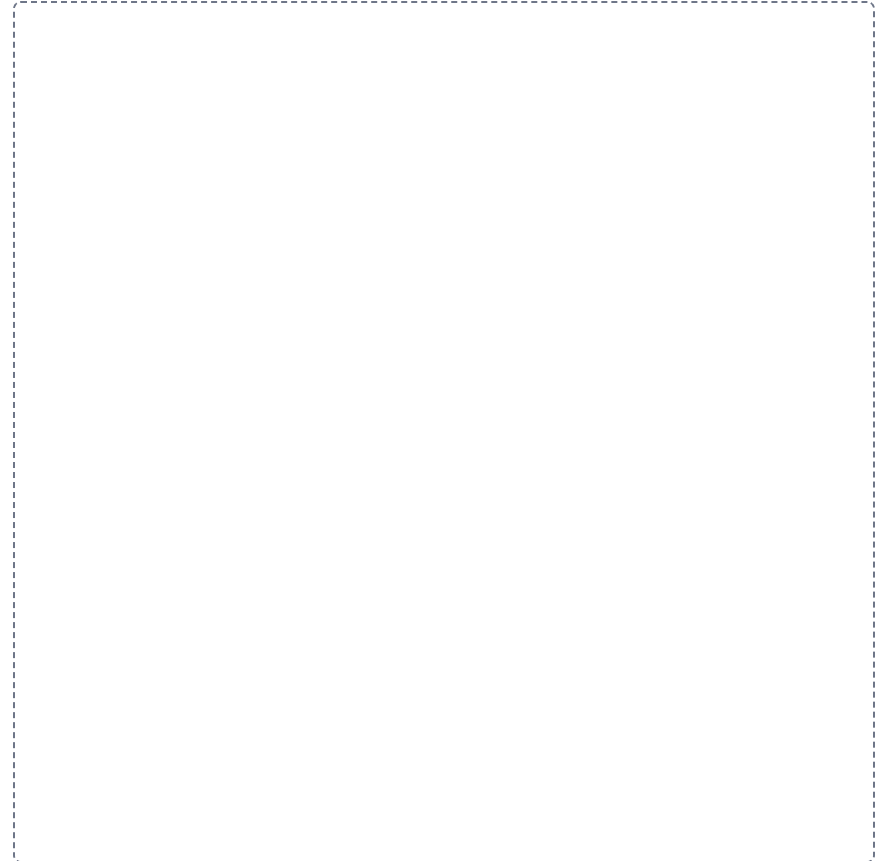
GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

**The five vessel types**

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

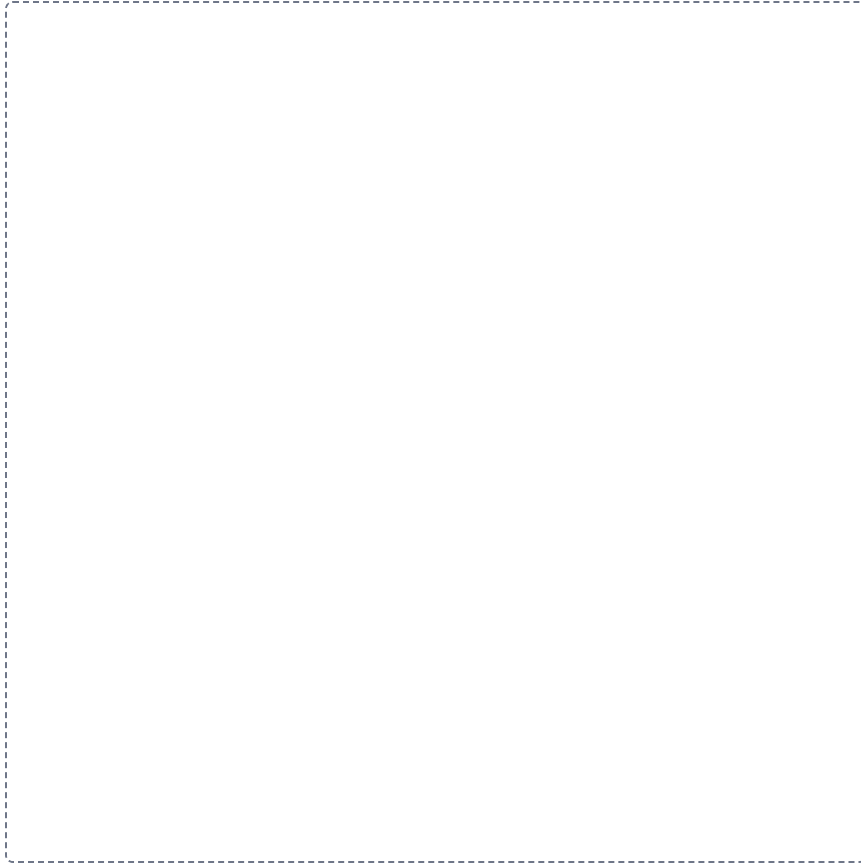


13-3

Artery against vein, on a slide

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

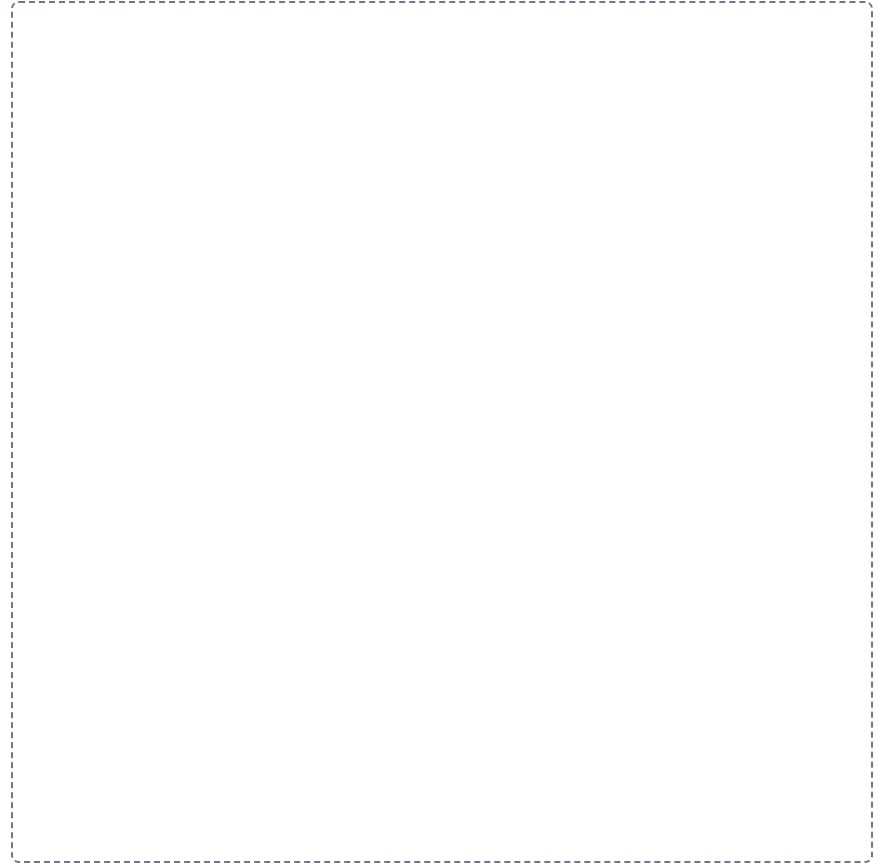


13-4

The three capillary types

GRID

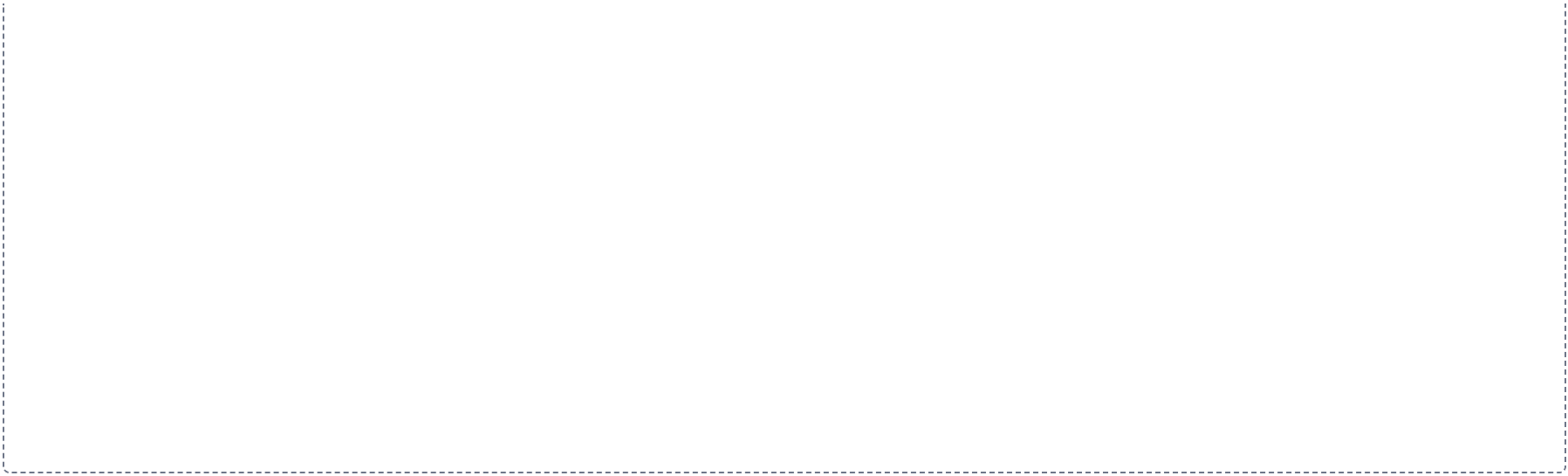
Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



Fetal circulation, placenta to placenta

CHAIN

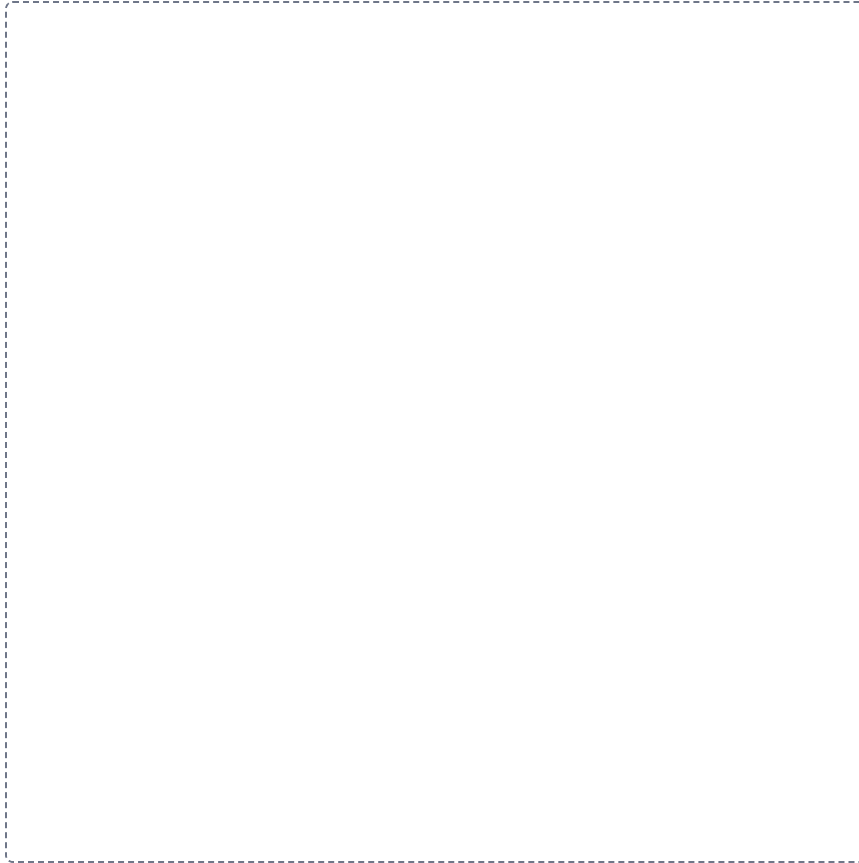
Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



Fetal shunt to adult remnant

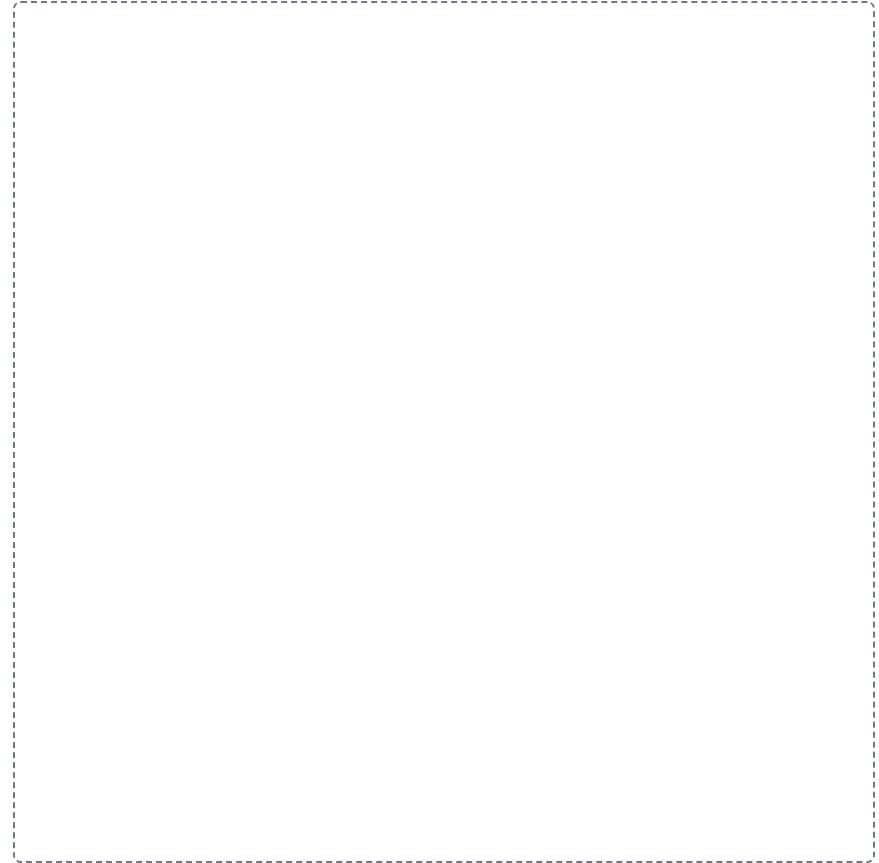
GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

**Blood, composition**

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



The five leukocytes

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

